

GREENING IT AND THE DATA CENTRES

Some very effective ways to maximise energy efficiency and minimise the environmental impact of computing equipment used in data centres



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Bill Gates had predicted the presence of a PC in every home long ago. At that time he was considered to be promoting an unlikely scenario with the aim of boosting Microsoft's profits. But if we look at the reality of today, how right he was? Not just commercial organisations, however small, but even homes are heavily reliant upon IT.

Though computers have brought a wealth of knowledge to our fingertips through the Internet, improved communication and reduced the amount of paper consumed, these are also harmful to us and our environment. They

are made of toxic materials like lead, chromium, cadmium and mercury, which can leach into the environment when the computers are buried into landfills and left in the open at the end of their life. Also, computers consume a lot of energy.

Green computing, or green IT, aims to overcome these problems.

What's green IT

At home, there is little we can do about our use of IT, other than not leaving our PCs switched on unnecessarily. But for organisations, there is a massive scope for recycling of their products and reducing their energy use. They can improve their public image, and hence profits, by adopting a green

approach to IT.

Benefits of green IT for businesses are:

Money saving. Green practices can save money. The U.S. Environmental Protection Agency (EPA) recommends shutting down your computer if it has to be inactive for more than a day. According to Microsoft, turning off your machine every night or putting it into hibernation mode can save \$90 per computer over the course of a year.

Customer preferences. Many companies make a virtue of ethics, building advertising campaigns around environmental awareness in the belief that customers prefer dealing with socially responsible businesses. Ben & Jerry's ice cream is a prominent example.

Traditional data centre

Traditionally, a data centre layout has rows of racks facing the same direction. The racks are arranged with just enough space in front and behind to service the equipment. The racks are 61cm (2 feet) wide and 91cm (3 feet) deep with an aisle in front and another at the back for service access. The aisles are 91cm wide to allow installation of servers into the racks. The reasoning here was that a 91cm deep rack could hold an equipment only 91cm deep or less.

The servers are physically bulky and not all were designed to be rack-mounted, leading to many servers sitting on shelves in the racks. Most x86-based servers are powered with 110V from power strips mounted in the racks or run directly from outlets located under the raised floor.